Understanding Sub-Seasonal Extremes in Semi-Arid Transition Zones: Using Machine-Learning to Enhance Predictability Signals for Early Action Decision-Making [STEP-AHEAD]

Fernanda Pino Delgado (Sup: Rosalind Cornforth[1], Ted Shepherd[1], Katherine Egan[2], Martín Jacques-Coper[3], Celia Petty[1], Abdou Ali[4])

[1]University of Reading · [2]ECMWF · [3]University of Concepción · [4]AGRHYMET

Theme 1: Predicting the Earth System up to the sub-seasonal range; **Cross-cutting theme:** Maximising the societal benefits of extended-range numerical weather predictions

Introduction

The project explores sources of predictability for extreme weather events 3-4 weeks ahead, with the aim of improving subseasonal forecasts. It focuses on semi-arid transition zones (Sahel and Atacama), as these regions are highly exposed to extremes and they also hold large areas of agricultural land, which livelihoods and food security depend on. The approach to this will be hybrid, combining physical-process understanding with machine learning to extract weak signals on the subseasonal time scale.

Objectives

1. Investigate predictable signals from physical-process understanding, at the subseasonal scale.
2. Apply machine learning methods to extract weak signals, to calibrate the forecasts and to explore improvements to the probabilistic interpretation of ensemble forecasts, at the subseasonal scale.
3. Assess how improved predictions can support early-action decision-making for food security and adaptive social protection.

